

What “Scenario” Means in IoT-23?

The IoT-23 dataset is organized into 23 separate scenario folders. Each scenario is basically a different recording session of network traffic.

- It contains traffic from one or more IoT devices.
- It may include benign traffic, malicious traffic, or both.
- Each scenario represents a specific attack case or setup.

#	Name of Dataset	Duration (hrs)	#Packets	#ZeekFlows	Pcap Size	Name
1	CTU-IoT-Malware-Capture-34-1	24	233,000	23,146	121 MB	Mirai
2	CTU-IoT-Malware-Capture-43-1	1	82,000,000	67,321,810	6 GB	Mirai
3	CTU-IoT-Malware-Capture-44-1	2	1,309,000	238	1.7 GB	Mirai
4	CTU-IoT-Malware-Capture-49-1	8	18,000,000	5,410,562	1.3 GB	Mirai
5	CTU-IoT-Malware-Capture-52-1	24	64,000,000	19,781,379	4.6 GB	Mirai
6	CTU-IoT-Malware-Capture-20-1	24	50,000	3,210	3.9 MB	Torii
7	CTU-IoT-Malware-Capture-21-1	24	50,000	3,287	3.9 MB	Torii
8	CTU-IoT-Malware-Capture-42-1	8	24,000	4,427	2.8 MB	Trojan
9	CTU-IoT-Malware-Capture-60-1	24	271,000,000	3,581,029	21 GB	Gagfyt
10	CTU-IoT-Malware-Capture-17-1	24	109,000,000	54,659,864	7.8 GB	Kenjiro
11	CTU-IoT-Malware-Capture-36-1	24	13,000,000	13,645,107	992 MB	Okiru
12	CTU-IoT-Malware-Capture-33-1	24	54,000,000	54,454,592	3.9 GB	Kenjiro
13	CTU-IoT-Malware-Capture-8-1	24	23,000	10,404	2.1 MB	Hakai
14	CTU-IoT-Malware-Capture-35-1	24	46,000,000	10,447,796	3.6G	Mirai
15	CTU-IoT-Malware-Capture-48-1	24	13,000,000	3,394,347	1.2G	Mirai
16	CTU-IoT-Malware-Capture-39-1	7	73,000,000	73,568,982	5.3GB	IRCBot
17	CTU-IoT-Malware-Capture-7-1	24	11,000,000	11,454,723	897 MB	Linux,Mirai
18	CTU-IoT-Malware-Capture-9-1	24	6,437,000	6,378,294	472 MB	Linux.Hajime
19	CTU-IoT-Malware-Capture-3-1	36	496,000	156,104	56 MB	Muhstik
20	CTU-IoT-Malware-Capture-1-1	112	1,686,000	1,008,749	140 MB	Hide and Seek

Table 1: Summary of the Malicious IoT Scenarios

Each folder has:

- Raw PCAP files (network captures)
- Pre-processed CSV files with features and labels (benign vs. attack)

When we say “choose a scenario,” it means select one folder and work with its corresponding CSV file for your analysis.

Why Use Two Different Scenarios?

The primary objective of Question 4 is to assess whether your model can identify attacks on data it has never encountered before, particularly from a different IoT device or attack type.

This mimics a real-world problem:

- You train your system using traffic from Device A (like a smart camera infected with Mirai).
- Later, a different device, like a smart plug or IoT router, gets infected, and you want to see if your model still works.

If the model only works well on the training scenario but fails on new scenarios, it means the model is overfitted and cannot generalize about unseen devices or attack behaviors.